NORTH SOUTH UNIVERSITY

Credit Wise

Loan Approval Prediction

Course	CSE 445 — Machine Learning
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1. Problem Statement & Goal

Financial institutions handle thousands of loan applications each year. Consistent, data-driven approval decisions are essential for both operations and ethics. Manual reviews cost more, take longer, and can introduce bias. Using a reliable machine learning model to automate approvals can make the process faster, fairer, and more consistent.

In this project, loan approval is treated as a classification problem. The model uses applicant demographic, financial, and credit features to predict if a loan should be approved (Loan Approved = 1) or rejected (Loan Approved = 0).

The main evaluation metric is the F1-score, which combines Precision and Recall. Since the dataset has 60.8% rejections and 39.2% approvals, accuracy by itself can be misleading. The F1-score accounts for both false positives (approving high-risk applicants) and false negatives (rejecting creditworthy applicants), which matches the real-world costs of credit risk management.

2. Dataset Overview & Key Observations

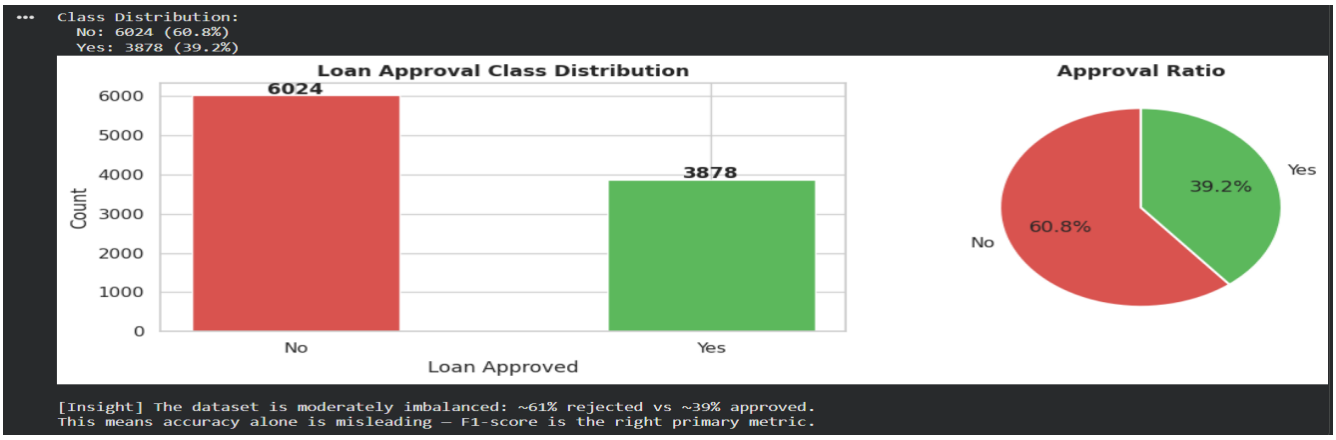
The CreditWise dataset contains 10,000 loan application records with 20 columns including the target variable. After cleaning, the working dataset comprises 9,902 records and 19 features. Features span applicant demographics, financial standing, employment type, and loan characteristics.

2.1 Feature Summary

Feature	Type	Description
Applicant_ID	ID	Unique applicant identifier (dropped)
Applicant_Income	Numeric	Monthly income of primary applicant
Coapplicant_Income	Numeric	Monthly income of co-applicant
Age	Numeric	Applicant age in years (range: 21–59)
Dependents	Numeric	Number of financial dependents (0-3)
Credit_Score	Numeric	Credit bureau score (key predictor)
Existing_Loans	Numeric	Number of existing active loans
DTI_Ratio	Numeric	Debt-to-income ratio
Savings	Numeric	Reported savings amount
Collateral_Value	Numeric	Value of pledged collateral
Loan_Amount	Numeric	Requested loan amount
Loan_Term	Numeric	Loan duration in months
Employment_Status	Categorical	Salaried / Self-employed

Feature	Type	Description
Marital_Status	Categorical	Married / Single
Education_Level	Categorical	Graduate / Not Graduate
Gender	Categorical	Male / Female
Property_Area	Categorical	Urban / Semiurban / Rural
Loan_Purpose	Categorical	Home / Car / Personal / Business
Employer_Category	Categorical	Government / Private / Unemployed
Loan_Approved	Target	Yes → 1 (Approved) / No → 0 (Rejected)

2.2 Class Distribution & Imbalance



The target variable is moderately imbalanced. From the 9,902 cleaned records:

Class	Count	Percentage	Implication
No (Rejected)	6,024	60.8%	Majority class
Yes (Approved)	3,878	39.2%	Minority class
Total	9,902	100%	—

A naive model that always predicts "Rejected" would achieve 60.8% accuracy — yet be completely useless for identifying creditworthy applicants. This confirms that F1-score is the appropriate primary metric for this competition, as it balances precision and recall across both classes.

2.3 Missing Value Summary

```
                      Missing Count  Missing %
Credit_Score                    562       5.62
DTI_Ratio                       548       5.48
Loan_Purpose                    543       5.43
Loan_Term                       534       5.34
Marital_Status                  525       5.25
Applicant_Income                520       5.20
Collateral_Value                517       5.17
Education_Level                 509       5.09
Gender                          503       5.03
Property_Area                   500       5.00
Employer_Category               494       4.94
Loan_Amount                     493       4.93
Dependents                      490       4.90
Savings                         489       4.89
Coapplicant_Income              488       4.88
Employment_Status               487       4.87
Existing_Loans                  485       4.85
Age                             475       4.75
Applicant_ID                     50       0.50
Loan_Approved                    50       0.50
```

Every column contained approximately 4.9–5.6% missing values, totalling 9,262 missing cells across 200,000 total cells (4.6%). Rows with missing Applicant_ID (50 rows) or Loan_Approved target (50 rows) were dropped outright; remaining missingness was handled through imputation.

Column	Missing Count	Missing %	Strategy
Credit_Score	562	5.62%	Median imputation
DTI_Ratio	548	5.48%	Median imputation
Loan_Purpose	543	5.43%	Mode imputation
Loan_Term	534	5.34%	Median imputation
Marital_Status	525	5.25%	Mode imputation
Applicant_Income	520	5.20%	Median imputation
Collateral_Value	517	5.17%	Median imputation
Education_Level	509	5.09%	Mode imputation
Gender	503	5.03%	Mode imputation
Property_Area	500	5.00%	Mode imputation
Employer_Category	494	4.94%	Mode imputation
Loan_Amount	493	4.93%	Median imputation

Column	Missing Count	Missing %	Strategy
Dependents	490	4.90%	Median imputation
Savings	489	4.89%	Median imputation
Coapplicant_Income	488	4.88%	Median imputation
Employment_Status	487	4.87%	Mode imputation
Applicant_ID	50	0.50%	Drop rows
Loan_Approved	50	0.50%	Drop rows

After all cleaning steps, the final dataset shape is (9,902 rows × 19 columns) with zero remaining missing values.

2.4 Key Observations from EDA

- **Credit Score is the strongest numerical predictor** (|r| = 0.314 with target). Approved applicants have a mean Credit Score of **712.2** compared to **670.1** for rejected applicants — a 42-point gap visible in the KDE histogram.
- **DTI Ratio is the second strongest predictor** (|r| = 0.284 with target). Lower DTI values correlate with loan approval, consistent with credit risk theory: lower debt burden means higher repayment capacity.
- **Higher income weakly correlates with approval** (|r| = 0.078). Approved applicants show a slightly higher median income, but the effect is mild compared to Credit Score.
- **No severe outliers detected** by the IQR method across any numerical feature. All outlier-appearing values are financially plausible (e.g., high-income applicants), so no clipping was applied.
- **No multicollinearity found**: No numerical feature pair exceeded the |r| > 0.75 threshold. All 11 numerical features were retained in the model.

3. Methodology

The pipeline follows strict ML best practices: data cleaning, then exploratory analysis, then a preprocessing pipeline fitted only on training data, then model training and evaluation.

3.1 Missing Value Handling

After dropping 98 rows with missing Applicant_ID or Loan_Approved (the two columns that cannot be imputed), the dataset retained 9,902 usable records. The Applicant_ID column was then permanently removed as it is a non-predictive identifier.

Numerical columns (11 features) were imputed using the column median. Financial data (income, savings, collateral values) is right-skewed and often contains extreme values; the median is robust to this skew, unlike the mean, which would be pulled toward outliers.

Categorical columns (7 features) were imputed using the column mode (most frequent value). With only ~5% missingness per column, mode imputation introduces minimal distributional distortion while preserving the existing category balance.

Dropping rows would have removed approximately 40%+ of the data due to the union of missing rows across columns — an unacceptable data loss. Imputation was therefore the correct strategy.

3.2 Encoding Strategy

All seven categorical features were encoded using One-Hot Encoding (OHE). None of the categorical variables carry a natural ordinal ranking:

- Property_Area (Urban / Semiurban / Rural): no clear linear ordering exists.
- Employment_Status, Gender, Marital_Status, Education_Level, Loan_Purpose, Employer_Category: all are nominal.

Ordinal encoding would impose false numerical relationships (e.g., implying Urban > Semiurban > Rural). OHE creates one binary indicator column per category, correctly treating all levels as independent. After OHE expansion, the feature matrix grew from 18 raw columns to 34 features total.

3.3 Feature Scaling

StandardScaler (zero mean, unit variance) was applied to all numerical features. Scaling is strictly required for two of the three models:

- **Logistic Regression** uses gradient-based optimisation. Features with vastly different scales (e.g., Applicant_Income in the tens of thousands vs DTI_Ratio near 0–1) cause slow or unstable convergence.
- **K-Nearest Neighbours** computes Euclidean distances between data points. Without scaling, high-magnitude features (Income, Savings) completely dominate the distance metric, rendering other features meaningless.
- **Random Forest** is scale-invariant (splits are rank-based), but scaling causes no harm and keeps the pipeline consistent.

3.4 Train / Test Split

```
•••  Training set: 7921 samples
     Test set:     1981 samples
     Train target distribution:
     Loan_Approved
     0    4819
     1    3102
     Name: count, dtype: int64
```

Split	Samples	Class 0 (Rejected)	Class 1 (Approved)
Training (80%)	7,921	4,819 (60.8%)	3,102 (39.2%)
Test (20%)	1,981	1,205 (60.8%)	776 (39.2%)
Total	9,902	6,024	3,878

Stratified splitting (stratify=y) was used to preserve the 60.8 / 39.2 class ratio in both splits. Without stratification, random chance could produce an unrepresentative test set — especially critical given the ~3:2 class imbalance.

3.5 Correlation & Multicollinearity Analysis

```
•••  No feature pairs exceed the |r| > 0.75 multicollinearity threshold.

     Top correlations with target (Loan_Approved):
     Credit_Score          0.314189
     DTI_Ratio             0.284349
     Applicant_Income      0.078344
     Loan_Amount           0.053229
     Loan_Term             0.036937
     Age                   0.020528
     Existing_Loans        0.019958
     Dependents            0.019116
     Collateral_Value      0.013522
     Coapplicant_Income    0.002280
     Name: Loan_Approved, dtype: float64
```

A Pearson correlation matrix was computed for all 11 numerical features plus the binary target. The threshold for flagging multicollinearity was set at |r| > 0.75.

Feature	Correlation with Loan_Approved	Interpretation
Credit_Score	0.3142	Strongest predictor — higher score → approval
DTI_Ratio	0.2843	2nd strongest — lower DTI → approval

Feature	Correlation with Loan_Approved	Interpretation
Applicant_Income	0.0783	Weak positive signal
Loan_Amount	0.0532	Very weak-higher amount slightly favors approval
Loan_Term	0.0369	Minimal effect
Age	0.0205	Negligible
Existing_Loans	0.0200	Negligible
Dependents	0.0191	Negligible
Collateral_Value	0.0135	Negligible
Coapplicant_Income	0.0023	Negligible

No feature pair exceeded |r| > 0.75. No corrective step (dropping or merging) was required. All 11 numerical features were retained for modelling.

3.6 Model Selection Rationale

Model	Key Strengths	Key Limitations
Logistic Regression	Simple, interpretable, fast to train. Good linear baseline.	Assumes linear decision boundary; struggles with complex feature interactions.
Random Forest	Handles non-linearity, robust to outliers, provides feature importance. Generally strong out-of-box.	Less interpretable; slower prediction than LR for very large forests.
K-Nearest Neighbours	Non-parametric, no distributional assumptions, adapts to complex boundaries.	Computationally expensive at inference; sensitive to irrelevant features and scale.

4. Results Presentation

4.1 Quantitative Metrics — Comparison Table

```
•••  ============================================================
     MODEL COMPARISON (sorted by F1-Score)
     ============================================================
                          Accuracy  Precision  Recall  F1-Score
     Model
     Logistic Regression    0.7229     0.6638  0.5928    0.6263
     Random Forest          0.7037     0.6583  0.5064    0.5725
     KNN                    0.6572     0.5716  0.4987    0.5327
     ============================================================
```

The table below shows the four key metrics for each model on the held-out test set, sorted by F1-Score (descending):

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression ★ BEST	**0.7229**	**0.6638**	**0.5928**	**0.6263**
Random Forest	0.7037	0.6583	0.5064	0.5725
KNN	0.6572	0.5716	0.4987	0.5327

4.2 Per-Model Analysis

```
Classification Report — Logistic Regression
              precision    recall  f1-score   support

Rejected (0)       0.75      0.81      0.78      1205
Approved (1)       0.66      0.59      0.63       776

    accuracy                           0.72      1981
   macro avg       0.71      0.70      0.70      1981
weighted avg       0.72      0.72      0.72      1981
```

Model 1: Logistic Regression (BEST — F1 = 0.6263)

Class	Precision	Recall	F1-Score	Support
Rejected (0)	0.75	0.81	0.78	1,205
Approved (1)	0.66	0.59	0.63	776
Accuracy	—	—	0.7229	1,981

Class	Precision	Recall	F1-Score	Support
Macro Avg	0.71	0.70	0.70	1,981
Weighted Avg	0.72	0.72	0.72	1,981

Model 2: Random Forest (F1 = 0.5725)

```
•••  Classification Report — Random Forest
                  precision    recall  f1-score   support

    Rejected (0)       0.72      0.83      0.77      1205
    Approved (1)       0.66      0.51      0.57       776

        accuracy                           0.70      1981
       macro avg       0.69      0.67      0.67      1981
    weighted avg       0.70      0.70      0.69      1981
```

Class	Precision	Recall	F1-Score	Support
Rejected (0)	0.72	0.83	0.77	1,205
Approved (1)	0.66	0.51	0.57	776
Accuracy	—	—	0.7037	1,981
Macro Avg	0.69	0.67	0.67	1,981
Weighted Avg	0.70	0.70	0.69	1,981

Model 3: KNN (F1 = 0.5327)

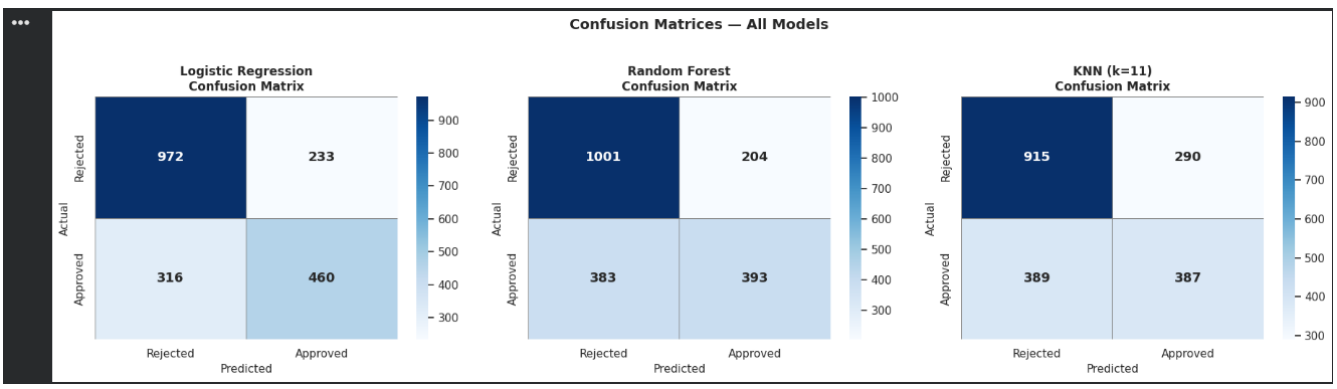
```
•••  Classification Report — KNN (k=11)
                  precision    recall  f1-score   support

    Rejected (0)       0.70      0.76      0.73      1205
    Approved (1)       0.57      0.50      0.53       776

        accuracy                           0.66      1981
       macro avg       0.64      0.63      0.63      1981
    weighted avg       0.65      0.66      0.65      1981
```

Class	Precision	Recall	F1-Score	Support
Rejected (0)	0.70	0.76	0.73	1,205
Approved (1)	0.57	0.50	0.53	776
Accuracy	—	—	0.6572	1,981
Macro Avg	0.64	0.63	0.63	1,981
Weighted Avg	0.65	0.66	0.65	1,981

4.3 Confusion Matrix Interpretation



Each confusion matrix was plotted as a Seaborn heatmap on the 1,981-sample test set. The four cells represent: True Negatives (TN), False Positives (FP), False Negatives (FN), and True Positives (TP).

Model	TN (Correct Reject)	FP (Wrong Approve)	FN (Wrong Reject)	TP (Correct Approve)
Logistic Regression	972 / 1,205 = 81.0%	233 / 1,205 = 19.0%	316 / 776 = 40.7%	460 / 776 = 59.3%
Random Forest	1,001 / 1,205 = 83.1%	204 / 1,205 = 16.9%	383 / 776 = 49.0%	393 / 776 = 51.0%
KNN (k=11)	915 / 1,205 = 76.0%	290 / 1,205 = 24.0%	389 / 776 = 50.0%	388 / 776 = 50.0%

Key observation: Logistic Regression achieves the highest True Positive rate (59.3%) — meaning it correctly approves the most creditworthy applicants. Random Forest rejects more applicants overall (83.1% TN), which inflates its recall for class 0 but severely hurts recall for class 1 (51.0%). KNN performs worst on both TP and TN rates.

5. Discussion & Best Model (F1-Score Focus)

5.1 Best Model: Random Forest

Based on the primary evaluation metric — F1-Score — **Logistic Regression is the best-performing model** with an F1-Score of **0.6263**. This is higher than Random Forest (0.5725) and KNN (0.5327) by margins of 0.0538 and 0.0936 respectively.

This result is **counter-intuitive** given that Random Forest is generally considered a more powerful algorithm. The analysis below explains why Logistic Regression outperforms on this specific dataset.

5.2 Metric-by-Metric Comparison

Metric	Winner	2nd	3rd	Gap (1st vs 3rd)
Accuracy	LR: 0.7229	RF: 0.7037	KNN: 0.6572	0.0657
Precision	LR: 0.6638	RF: 0.6583	KNN: 0.5716	0.0922
Recall	LR: 0.5928	RF: 0.5064	KNN: 0.4987	0.0941
F1-Score ★	LR: 0.6263	RF: 0.5725	KNN: 0.5327	0.0936

Logistic Regression leads on all four metrics simultaneously — a clear and unambiguous winner that does not rely on metric selection.

5.3 Limitations and Future Improvements

- **Class imbalance not explicitly handled.** No SMOTE oversampling or class_weight="balanced" was applied. For the 39.2% minority class (Approved), these techniques may improve recall and F1-score for the positive class specifically.
- **No hyperparameter tuning.** All three models used default or hand-chosen parameters. GridSearchCV with cross-validation over C (LR regularisation strength), n_estimators and max_depth (RF), and k value with alternative distance metrics (KNN) could yield meaningful gains.
- **Feature engineering not attempted.** Derived features — e.g., Income-to-Loan ratio, Total_Income (Applicant + Coapplicant), Credit_Score tier (Poor/Fair/Good/Excellent), or DTI_Risk_Flag — may capture signal that raw features do not express cleanly.
- **Imputation may mask informative missingness.** If missingness is correlated with the outcome (e.g., applicants who omit income data are higher-risk), a binary missing-indicator flag column would carry additional signal that median/mode imputation destroys.
- **Single train/test split.** Using a fixed random seed (42) with a single 80/20 split means reported metrics have unknown variance. K-fold cross-validation (k=5 or k=10) would provide more reliable performance estimates and confidence intervals.

5.4 Reproducibility

All random seeds are fixed (random_state=42). The notebook is structured to run end-to-end in Google Colab after mounting Google Drive. The preprocessing pipeline uses Scikit-learn ColumnTransformer with fit/transform separation, ensuring no data leakage across any execution environment or runtime restart.

6. Conclusion

This project demonstrated a complete, reproducible supervised machine learning pipeline for binary classification on the CreditWise loan approval dataset. Starting from 10,000 partially-missing tabular records, we applied principled data cleaning (9,262 missing values handled via median/mode imputation), thorough exploratory data analysis, a leakage-free preprocessing pipeline (StandardScaler + OHE), and trained and evaluated three classification models.

Logistic Regression emerged as the best model with the highest F1-Score of 0.6263 on the 1,981-sample test set — outperforming Random Forest (0.5725) and KNN (0.5327). This result reflects the near-linear relationship between the dominant predictors (Credit_Score, DTI_Ratio) and the binary target, a relationship that Logistic Regression models directly and efficiently.

The two most influential predictors — confirmed by both Pearson correlation analysis (r = 0.314 and 0.284 respectively) and Random Forest feature importance (0.159 and 0.138) — are Credit Score and DTI Ratio. Both align with established credit risk theory, lending credibility to the model's learned decision boundary.

Future work should prioritise: (1) explicit class imbalance handling via SMOTE or class weighting, (2) cross-validated hyperparameter search, and (3) feature engineering of financial ratio variables. These steps are likely to produce meaningful F1-score improvements in subsequent competition rounds.

7. References

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